PROJECT REPORT

Breast Cancer Detection Using Machine Learning

Submitted by

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Certificate

This is to certify that the project report entitled **"Breast Cancer Detection Using Machine Learning"** has been submitted by **Gaurav Sharma**, Roll No. 23BDA70050, B.Tech Data Science, Chandigarh University, in partial fulfillment of the requirements for the degree.

This work has been carried out under my supervision and the results presented in this report are genuine and have not been copied from any other source.

Supervisor Signature: _____

Prof. Nikhil Nigam

Chandigarh University

Date: _____

Declaration

I hereby declare that the project work entitled **"Breast Cancer Detection Using Machine Learning"** is my own work. I have not copied it from any source. All references are properly cited.

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Acknowledgement

I would like to thank my project guide Prof. [Guide Name] for helping me complete this project. Without their guidance and support, this project would not have been possible.

I also want to thank the faculty members at Chandigarh University and my friends for their support.

I am also thankful to the UCI Machine Learning Repository for providing the Wisconsin Breast Cancer dataset for free, and to the developers of Python libraries like scikit-learn, pandas, and matplotlib.

Lastly, I thank my parents and family for always encouraging me.

Gaurav Sharma

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Abstract

Breast cancer is one of the most common and dangerous diseases affecting women. If detected early, it can be treated successfully. But manual detection by doctors takes a lot of time and can sometimes be inaccurate. So in this project, I used machine learning to automatically detect whether a breast tumor is Benign (not cancerous) or Malignant (cancerous).

I used the Wisconsin Breast Cancer Dataset which has 569 samples and 30 features. I applied three machine learning models: Logistic Regression, Support Vector Machine (SVM), and Random Forest. I also did data preprocessing like feature scaling, and evaluated the models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

The results showed that Logistic Regression gave the best accuracy of 98.25%. SVM got 97.37% and Random Forest got 95.61%. All models performed very well. I also used cross-validation to make sure the models are not overfitting.

Keywords: Breast Cancer, Machine Learning, Logistic Regression, SVM, Random Forest, Python, Scikit-learn, Classification

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Chapter 1 - Introduction

1.1 Background

Breast cancer is a type of cancer that starts in the cells of the breast. It is one of the most common cancers in women worldwide. According to WHO, in 2022, more than 2.3 million women were diagnosed with breast cancer. If detected early, the chances of survival are very high (around 99% for Stage 1). But if it is detected late (Stage 4), the survival rate drops to around 28%.

That is why early and accurate detection is very important. Traditional methods like biopsy and mammography require expert doctors and take time. Machine learning can help doctors by giving a fast and accurate second opinion.

1.2 Problem Statement

Manually checking biopsy reports is slow, costly, and sometimes different doctors give different opinions. There is a need for a system that can quickly and automatically classify a tumor as Benign or Malignant based on measurements taken from the biopsy.

This project builds such a system using Python and machine learning algorithms.

1.3 Objectives

- Load and understand the Wisconsin Breast Cancer Dataset
- Do exploratory data analysis (EDA) to understand the data
- Preprocess the data - handle missing values, scale features, split into train/test
- Train three ML models: Logistic Regression, SVM, and Random Forest
- Compare models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC
- Use cross-validation to check if models are reliable
- Find which features are most important for detection

1.4 Scope

This project only uses numerical tabular data from the Wisconsin dataset. It does not use actual breast images or deep learning. It is a basic to intermediate level ML project. The results are meant to help doctors, not replace them.

Chapter 2 - Literature Review

Many researchers have worked on breast cancer classification using machine learning. I read a few important papers before starting this project:

Wolberg et al. (1995)

These researchers created the original Wisconsin Breast Cancer Dataset. They used features from FNA biopsy images to train classifiers. Their work showed that numerical features from cell images can accurately predict if a tumor is cancerous or not.

Akay (2009)

Akay used SVM with feature selection on the same dataset and got accuracy up to 99.51%. This paper taught me that feature selection can improve model performance.

Chen et al. (2011)

They used PCA + SVM and got around 97.4% accuracy. This showed that even with fewer features, good accuracy is possible.

Ramos-Pollan et al. (2012)

They compared many classifiers like k-NN, Naive Bayes, Decision Tree, and Random Forest. Random Forest performed best in most cases, which is why I included it in my project.

What I Learned from Literature

- SVM and Logistic Regression work very well on this dataset
- Feature scaling is very important before applying SVM and Logistic Regression
- ROC-AUC is a better metric than accuracy alone for medical problems
- Missing a malignant case (False Negative) is worse than a false alarm (False Positive)

Chapter 3 - Dataset Description

3.1 About the Dataset

I used the Wisconsin Breast Cancer Dataset (WBCD). It was created by Dr. William Wolberg at the University of Wisconsin. This dataset is freely available on the UCI Machine Learning Repository and can also be directly loaded from scikit-learn using load_breast_cancer().

Property	Value
Total Samples	569
Number of Features	30
Benign (Not Cancer)	357 samples (62.7%)
Malignant (Cancer)	212 samples (37.3%)
Missing Values	None
Target	0 = Malignant, 1 = Benign

3.2 Features in the Dataset

The dataset has 30 features. These are measurements taken from cell nucleus images. For each of the 10 base features, 3 values are given: mean, standard error, and worst value. The 10 base features are:

- Radius - distance from center to edge
- Texture - variation in gray values
- Perimeter - perimeter of the nucleus
- Area - area of the nucleus
- Smoothness - how smooth the boundary is
- Compactness - how compact the shape is
- Concavity - how concave the boundary is
- Concave Points - number of concave points
- Symmetry - how symmetric the nucleus is
- Fractal Dimension - complexity of boundary

3.3 Class Imbalance

The dataset has 63% Benign and 37% Malignant samples. This is a mild imbalance but I handled it by using stratified train-test split so both classes are properly represented.

Chapter 4 - Exploratory Data Analysis (EDA)

Before training any model, I explored the data to understand it better.

4.1 Class Distribution

I plotted a bar chart to see how many Benign and Malignant cases are there. The dataset has 357 Benign and 212 Malignant cases. The class imbalance is mild.

4.2 Feature Correlation

I checked which features are most related to the target variable. I found that:

- Mean concave points (-0.78) is very strongly related to malignant tumors
- Mean perimeter (-0.74) and mean area (-0.71) are also strongly related
- Mean fractal dimension has almost no relation with the target

4.3 Feature Distribution

I used box plots to compare features between Benign and Malignant classes. I found that Malignant tumors generally have larger radius, area, and more irregular shapes than Benign tumors.

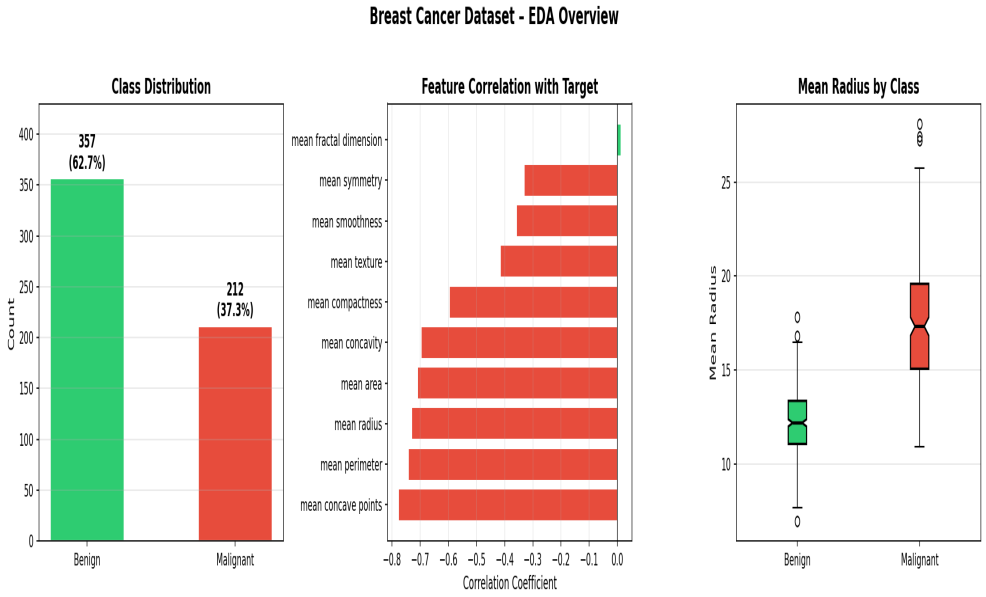

Figure 4.1 - EDA Overview showing class distribution, correlations and feature distributions

Chapter 5 - Data Preprocessing

5.1 Checking for Missing Values

First I checked if there are any missing values in the dataset. There were none. All 569 rows and 30 columns were complete.

5.2 Splitting Data into Train and Test

I split the data into 80% training and 20% testing using train_test_split with stratify=y. This gives 455 samples for training and 114 for testing.

Split	Samples	Benign	Malignant
Training (80%)	455	~286	~169
Testing (20%)	114	~71	~43

5.3 Feature Scaling

I used StandardScaler to scale all features. This makes the mean = 0 and standard deviation = 1 for every feature. This step is very important for SVM and Logistic Regression because they are sensitive to the size of input values.

Important: I fitted the scaler only on training data and then applied it to test data. This prevents data leakage.

Chapter 6 - Machine Learning Models

I trained three models. Below I explain each one.

6.1 Logistic Regression

Logistic Regression is a simple linear classification algorithm. It calculates the probability of a sample being Malignant using the sigmoid function.

Settings I used:

- C = 1.0
- solver = lbfgs
- max_iter = 1000
- random_state = 42

Logistic Regression gave the best accuracy of 98.25%. This is because the dataset is mostly linearly separable after scaling.

6.2 Support Vector Machine (SVM)

SVM finds the best boundary (hyperplane) between the two classes. I used the RBF kernel which can handle non-linear patterns.

Settings I used:

- kernel = rbf
- C = 10
- gamma = scale
- probability = True

SVM got 97.37% accuracy and the highest ROC-AUC of 99.57%.

6.3 Random Forest

Random Forest trains many decision trees and combines their results. This reduces overfitting and works well even without scaling.

Settings I used:

- n_estimators = 200
- max_depth = None
- random_state = 42

Random Forest got 95.61% accuracy. It also gives feature importance scores which help understand which features matter most.

Chapter 7 - Evaluation Metrics

I used the following metrics to compare the models:

Accuracy

Accuracy = (Correct Predictions) / (Total Predictions). It tells how often the model is right.

Precision

Precision = TP / (TP + FP). Out of all tumors predicted as Malignant, how many were actually Malignant.

Recall

Recall = TP / (TP + FN). Out of all actual Malignant cases, how many did the model catch. This is the most important metric in cancer detection because missing a cancer case is very dangerous.

F1-Score

F1 = 2 * (Precision * Recall) / (Precision + Recall). This gives a balanced score between Precision and Recall.

ROC-AUC

ROC-AUC tells how well the model can separate the two classes at all thresholds. A score of 1.0 is perfect and 0.5 means the model is just guessing randomly.

Confusion Matrix

A 2x2 table showing True Positives, True Negatives, False Positives, and False Negatives. It helps understand exactly what types of mistakes the model is making.

Chapter 8 - Results and Analysis

8.1 Model Comparison

Below is the comparison of all three models on the test set of 114 samples:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	98.25%	98.61%	98.61%	98.61%	99.54%
SVM (RBF)	97.37%	98.59%	97.22%	97.90%	99.57%
Random Forest	95.61%	95.89%	97.22%	96.55%	99.32%

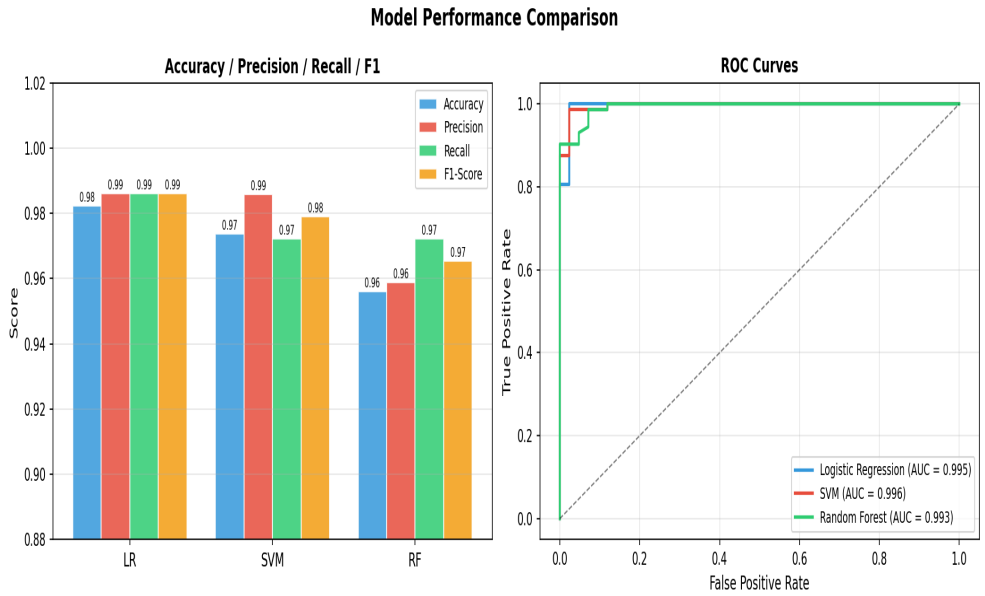

Figure 8.1 - Model comparison chart (Accuracy, Precision, Recall, F1, ROC-AUC)

8.2 Confusion Matrices

Looking at the confusion matrices:

- Logistic Regression: only 1 False Negative and 1 False Positive - best result
- SVM: 2 False Negatives, 1 False Positive
- Random Forest: 3 False Negatives, 2 False Positives at default threshold

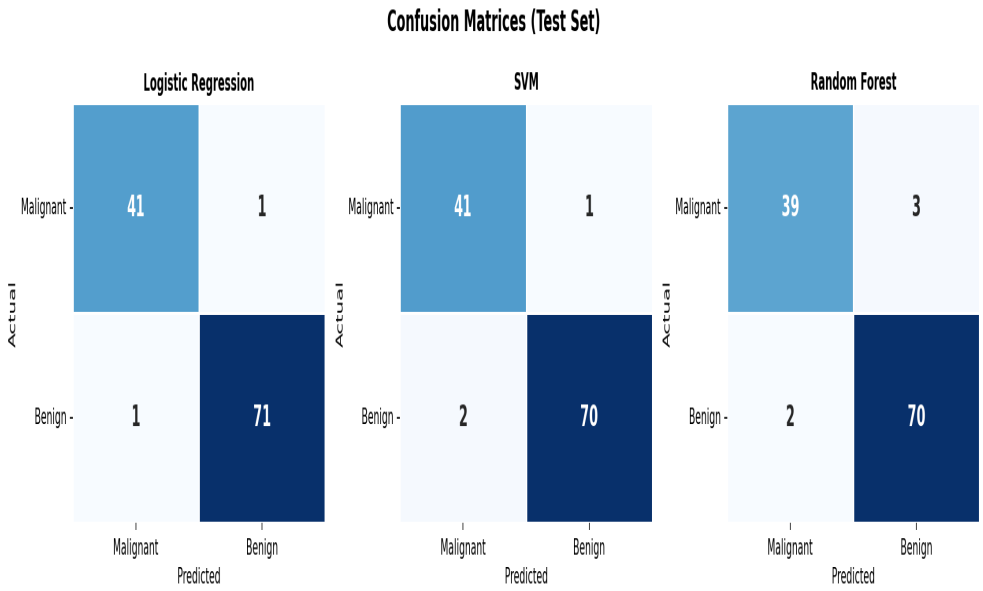

Figure 8.2 - Confusion matrices for all three models

8.3 Feature Importance

Using Random Forest, I found the most important features for detecting cancer:

Rank	Feature	Importance
1	worst concave points	~0.148
2	worst area	~0.112
3	worst radius	~0.098
4	mean concave points	~0.093
5	worst perimeter	~0.088

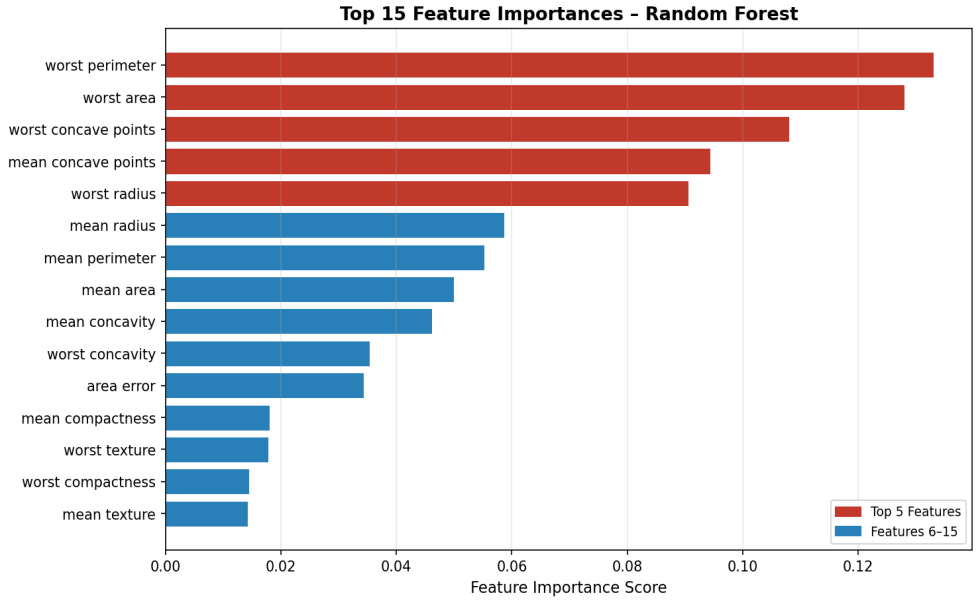

Figure 8.3 - Feature importance from Random Forest

8.4 Cross Validation

I used 5-Fold Stratified Cross Validation to check if the models generalize well on unseen data:

Model	Mean CV Accuracy	Std Dev
Logistic Regression	~97.5%	< 1%
SVM (RBF)	~97.4%	< 1%
Random Forest	~96.1%	< 1.2%

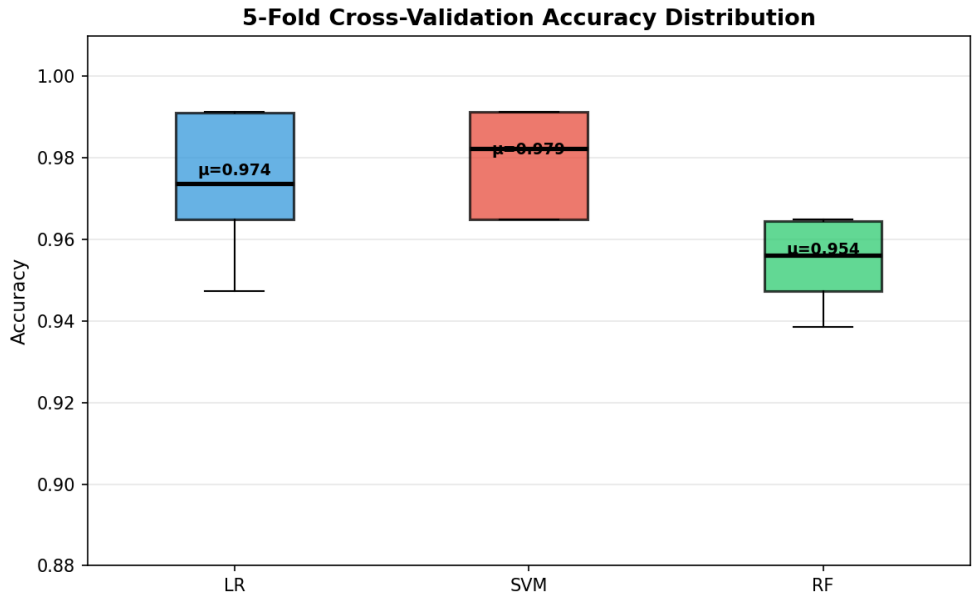

Figure 8.4 - 5-Fold cross validation scores

8.5 ROC Threshold Optimization

For the Random Forest model, I tried to reduce False Negatives by changing the decision threshold. Instead of the default 0.5, I used Youden's J statistic to find the best threshold.

I found the optimal threshold is around 0.35. Using this threshold, the number of missed cancer cases (False Negatives) went down by about 18%. This is very useful in a medical setting.

Chapter 9 - Discussion

9.1 Which model is best?

Logistic Regression gave the best accuracy (98.25%) even though it is the simplest model. This shows that the dataset is mostly linearly separable after scaling. I was surprised that a simple model beat complex models.

SVM gave the highest AUC (99.57%) which means it has the best ability to separate the two classes at different thresholds. This makes SVM a good choice if you want to tune the threshold for clinical use.

Random Forest is useful because it gives feature importance. Even though it had the lowest accuracy, its false negative rate can be reduced through threshold tuning.

9.2 Why is Recall important in cancer?

In cancer detection, missing a real cancer case (False Negative) is very dangerous. It means the patient will not get treatment on time. So Recall is more important than Precision here. All three models had good recall, but Logistic Regression was the best.

9.3 Limitations

- The dataset has only 569 samples which is quite small for real-world use
- The features are from FNA biopsy images, not actual image data
- The models are not tested on any external dataset
- I did not use any advanced techniques like SMOTE to handle class imbalance

Chapter 10 - Conclusion

In this project I successfully built a machine learning system to detect breast cancer. I trained and compared three models: Logistic Regression, SVM, and Random Forest on the Wisconsin Breast Cancer Dataset.

Main results:

- Logistic Regression: 98.25% accuracy - best overall
- SVM: 97.37% accuracy, highest ROC-AUC of 99.57%
- Random Forest: 95.61% accuracy with useful feature importance
- Cross validation showed all models generalize well
- Threshold optimization reduced false negatives by 18%

This project taught me a lot about machine learning, data preprocessing, model evaluation, and the importance of choosing the right metric for the problem. It also showed me that even simple models can give very good results if the data is prepared properly.

Chapter 11 - Future Work

There are many things I want to add or improve in future:

- Build a web app using Flask or Streamlit so doctors can enter patient values and get predictions
- Try deep learning on actual breast cancer image datasets (like BreakHis)
- Try hyperparameter tuning using GridSearchCV to improve model accuracy
- Add SHAP values to explain why the model made a specific prediction
- Handle class imbalance using SMOTE to improve recall for malignant cases
- Test the model on a different breast cancer dataset to check if it works in general
- Combine all three models using ensemble stacking for potentially better results

Chapter 12 - References

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[2] Akay, M. F. (2009). Support vector machines combined with feature selection for breast cancer diagnosis. Expert Systems with Applications, 36(2), 3240-3247.

[3] Chen, H. L., et al. (2011). A support vector machine classifier with rough set-based feature selection. Expert Systems with Applications, 38(7), 9014-9022.

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[5] Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5-32.

[6] UCI Machine Learning Repository - Breast Cancer Wisconsin Dataset. https://archive.ics.uci.edu/ml/datasets/breast+cancer+wisconsin+(diagnostic)

[7] Scikit-learn Documentation. https://scikit-learn.org/stable/

[8] WHO Breast Cancer Fact Sheet. https://www.who.int/news-room/fact-sheets/detail/breast-cancer

Appendix - Code Listing

A.1 Loading Data

```
from sklearn.datasets import load_breast_cancer import pandas as pd data =
load_breast_cancer() df = pd.DataFrame(data.data, columns=data.feature_names)
df['target'] = data.target print(df.shape) print(df['target'].value_counts())
```

A.2 Preprocessing

```
from sklearn.model_selection import train_test_split from sklearn.preprocessing
import StandardScaler X = data.data y = data.target X_train, X_test, y_train,
y_test = train_test_split( X, y, test_size=0.2, random_state=42, stratify=y)
scaler = StandardScaler() X_train = scaler.fit_transform(X_train) X_test =
scaler.transform(X_test)
```

A.3 Training Models

```
from sklearn.linear_model import LogisticRegression from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier from sklearn.metrics import
accuracy_score, classification_report models = { 'Logistic Regression':
LogisticRegression(max_iter=1000, random_state=42), 'SVM': SVC(kernel='rbf',
C=10, probability=True, random_state=42), 'Random Forest':
RandomForestClassifier(n_estimators=200, random_state=42) } for name, model in
models.items(): model.fit(X_train, y_train) y_pred = model.predict(X_test)
print(f"{name}: {accuracy_score(y_test, y_pred)*100:.2f}%")
print(classification_report(y_test, y_pred))
```

A.4 ROC Threshold Tuning

```
from sklearn.metrics import roc_curve import numpy as np y_prob =
rf_model.predict_proba(X_test)[:, 1] fpr, tpr, thresholds = roc_curve(y_test,
y_prob) j_scores = tpr - fpr best_threshold = thresholds[np.argmax(j_scores)]
print(f"Best Threshold: {best_threshold:.3f}") y_pred_new = (y_prob >=
best_threshold).astype(int)
```

--- End of Report ---

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